Sunil Madhow

San Diego, CA | smadhow@ucsd.edu | sunilmadhow.github.io | linkedin.com/in/sunil-madhow
github.com/SunilMadhow

Research Interests

I develop theoretically principled machine-learning algorithms that adapt to distribution shift, limited feedback, and other complexities of real-world data. My research spans statistical learning theory, online learning, reinforcement learning, and inference-time alignment.

Education

University of California, San Diego, Ph.D @ Halıcıoğlu Data Science Institute 2024-Present

University of California, Santa Barbara, MS in Computer Science 2022-23

- GPA: 3.96/4.0

University of California, Santa Barbara, BS in Computer Science 2019-22

- GPA (Major GPA): 3.93(3.98)/4.0

Selected Coursework:
Statistics: High-Dimensional Statistics, Mathematical Statistics, Stochastic Processes, Information Theory;
ML: Machine Learning, Statistical Learning Theory, Online Learning, Mathematical Deep Learning;
Math & TCS: Complexity Theory, Game Theory, Real Analysis, Complex Analysis, Randomized Algorithms, Spectral Graph Theory, Control Theory, Formal Logic, Post-Quantum Cryptography, Advanced Operating Systems.

Papers

S. Madhow, Y. Liang, N. Shroff, Y. Liang, Y.-X. Wang. Learnable Chernoff Baselines for Inference-Time Alignment. **To appear in NeurIPS 2026**.
S. Madhow, D. Baby, Y.-X. Wang. AKORN: Adaptive Knots Generated Online for Regression Splines. **ICML 2025**.
S. Madhow, D. Qiao, M. Yin, Y.-X. Wang. Rates for Offline Reinforcement Learning with Adaptively Collected Data. **L4DC 2025**.
S. Madhow, Y.-X. Wang. "Deep lifts": Neural networks can generalize even with the dumbest optimizer. **Under review for ICLR 2027**.
Y.-X. Wang, S. Madhow. Revisiting Off-Policy Evaluation in Stationary MDPs: When is temporal pooling helpful? **Under review for AISTATS 2027**.

Experience

Applied Science Intern II, Amazon Summer, 2026

- Designed and implemented RL algorithms for end-to-end training of multi-agent fraud classification pipeline at scale; the goal was to replace Claude queries with an in-house model.
- Improved ROC-AUC, PR-AUC, and reasoning–prediction consistency relative to a Claude Sonnet-based production baseline.
- Received return offer

Applied Science Intern II, Amazon Summer, 2025

- Research on Reinforcement Learning for optimizing inventory pricing/disposal under highly nonstationary market conditions.
- By ensembling RL agents via a bandit-based selector, boosted long-term free cash flow by **10**-**40**% across simulated markets (relative to best agent in ensemble); improved robustness/adaptivity under adversarial and/or random shocks to elasticity, demand, supply cost.
- Proved optimal theoretical rates for the ensembling algorithm under idealized assumptions; optimized implementation in Python (PyTorch) with parallelization.

Projects

Architecture-based Bounds for Generalization and Overfitting Under review, manuscript.

- Proved architecture-based generalization bounds of the form $\tilde{O}(\mathbf{Comp}(f) \times (\text{width} \times \text{depth})/n)$ for random interpolating deep neural networks with continuous weights. Here, $\mathbf{Comp}(f)$ measures the complexity of realizing the ground truth, f, with a neural network. width and depth are the dimensions of the neural network.
- Extended generalization analysis to the case of noisy data, proving that random interpolating networks achieve (at least) tempered overfitting with high probability.
- Proved that overfitting is benign when networks are sampled from the set of networks that fit below the intrinsic noise level of the data.

Learnable Chernoff Baselines for Inference-Time Alignment NeurIPS 2026

- Introduced fully black-box algorithm for (approximately) sampling from optimal reward-aligned distribution for any diffusion or language model.
- Derived end-to-end and finite-sample guarantees on TV sampling error for model-agnostic guidance. These bounds exhibit fine-grained dependence on regularity of underlying pretrained model.
- Reduced number of queries to pretrained model by 40% in large-scale language domain, and by $7\times$ in toy continuous domain, compared to Best-of-N alignment.
- To appear at *NeurIPS, 2026*.

AKORN: Adaptive Knots generated Online for RegressioN splines ICML 2025

- Designed a provably optimal *online* prediction algorithm for sequences coming from higher-order TV class.
- Proposed AKORN: the first provably optimal knot-selection algorithm via reduction to online prediction.
- AKORN is parameter-free but competes with SOTA (Trend Filtering) in empirical experiments
- Published in *International Conference on Machine Learning, 2025*

Offline Reinforcement Learning with Adaptively Collected Data L4DC 2025

- Derived bounds on sample complexity for offline policy learning and evaluation, including minimax optimal bounds.
- Key question: how to deal with an offline dataset where distribution can drift arbitrarily?
- Published in *Learning for Dynamics and Control Conference 2025*.

Programming Languages

Languages: Python, C, C++, OCaml, Java, Javascript
Packages: PyTorch, verl, NumPy, Pandas, Scikit-learn, Stable Baselines, React